Computational Environments and Toolchains

Topic 02 — Scientific Computing using Python

Lecture 02 — Numpy

Kieran Murphy and David Power

Department of Computing and Mathematics,
SETU (Waterford).
(kieran.murphy@setu.ie, david.power@setu.ie)

Autumn Semester, 2022

Outline

- matplotlib (2D and 3D plotting library)
- numpy (high performance matrix library)
- scipy (scientific computation library)

Outline

1. numpy 2
1.1 Introduction 3
1.2 Array Creation 4
1.3 Array Operations 7
1.4 Array Slicing 8
1.5 Array Broadcasting 10

What is NumPy?

NumPy

is an extension to the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large library of high-level mathematical functions to operate on these arrays.

- At the core of the NumPy package, is the ndarray object which encapsulates n-dimensional arrays of homogeneous data.
- Many operations performed using ndarray objects execute in compiled code for performance.
- The standard mathematical and scientific packages in Python use NumPy arrays.

Array Creation

array_creation .py

```
import numpy as np

# from a list
arr = np.array([[1, 2, 3], [4, 5, 6]])

# from sequences
np.arange(0, 10, 0.1)
np.linspace(0, 2*np.pi, 100)

# zeros & ones
np.zeros((5, 5))
np.ones((5, 5))

# random
np.random.random(size=(3, 4))
np.random.normal(loc=10., scale=3., size=(3, 4, 5))
```

Array IO

array_io.py

```
import numpy as np

# create an array, write to file, read from file
arr = np.array([[1, 2, 3], [4, 5, 6]])

# save to a text file
# creates a space delimited file by default
np.savetxt(fname='array_out.txt', X=arr)

# load text file
loaded_arr = np.loadtxt(fname='array_out.txt')

# verify that save->load worked
np.all(arr == loaded_arr)  # True
```

Array Attributes

Arrays are objects and so have attributes and methods.

array_attributes .py

```
import numpy as np

arr = np.arange(10).reshape((2, 5))

arr.ndim          # 2 number of dimensions
arr.shape         # (2, 5) shape of the array
arr.size          # 10 number of elements
arr.T             # transpose
arr.dtype         # data type of elements in the array
```

And many others. Explore in documentation or with TAB complete in ipython.

Array Operations & ufuncs

array_operations .py

```
arr1 = np.arange(10).reshape((2, 5))
arr2 = np.random.random((2, 5))

# element-wise for basic and boolean operations
#    +, -, *, /, **, np.log, <, >=, ==
#    arrays are upcast, resulting in float or boolean arrays
arr1 + arr2          # elementwise sum
arr1 * arr2          # elementwise multiplication

# operations in place
arr1 += arr2

# matrix product
np.dot(arr1, arr2)

# similarly numpy ufunc's operate element-wise
np.sin(arr1)
np.sqrt(arr1)
```

Default behaviour is element-wise

Array Slicing

More powerful than slicing in lists (see boolean indexing).

array_slice .py

```
arr = np.arange(20).reshape((4, 5))

# slicing (like lists for each dimension)
arr[0:4, 3:5]  # all rows and last two columns
arr[:4,  3:5]  # equivalent - can leave off start
arr[:, 3:]     # equivalent - can leave off end
arr[slice(None), slice(3, None)]  # equivalent - can use slice()

# integer indices
arr[[1, 2], :]                # rows one and two, all columns
arr[np.array([1, 2]), :]      # equivalent

# boolean indices
arr[[False, True, True, False], :]                # equivalent
arr[np.array([False, True, True, False]), :]      # equivalent
```

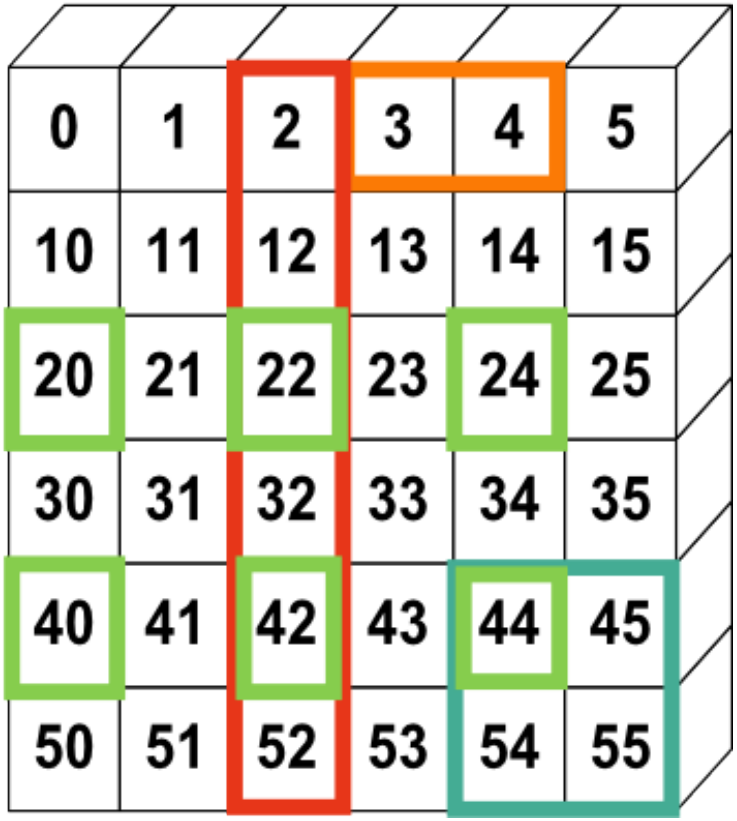
Array Slicing

```
>>> a[0,3:5]
array([3,4])

>>> a[4:,4:]
array([[44, 45],
       [54, 55]])

>>> a[:,2]
array([2,12,22,32,42,52])

>>> a[2::2,::2]
array([[20,22,24]
       [40,42,44]])
```



From Scipy Lecture Notes
(http://www.scipy-lectures.org/index.html)

Array Broadcasting & Vectorisation

Broadcasting allows us to operate on arrays of different shapes by 'copying' smaller arrays when possible. This allows us to write more efficient and readable code (with fewer for loops).

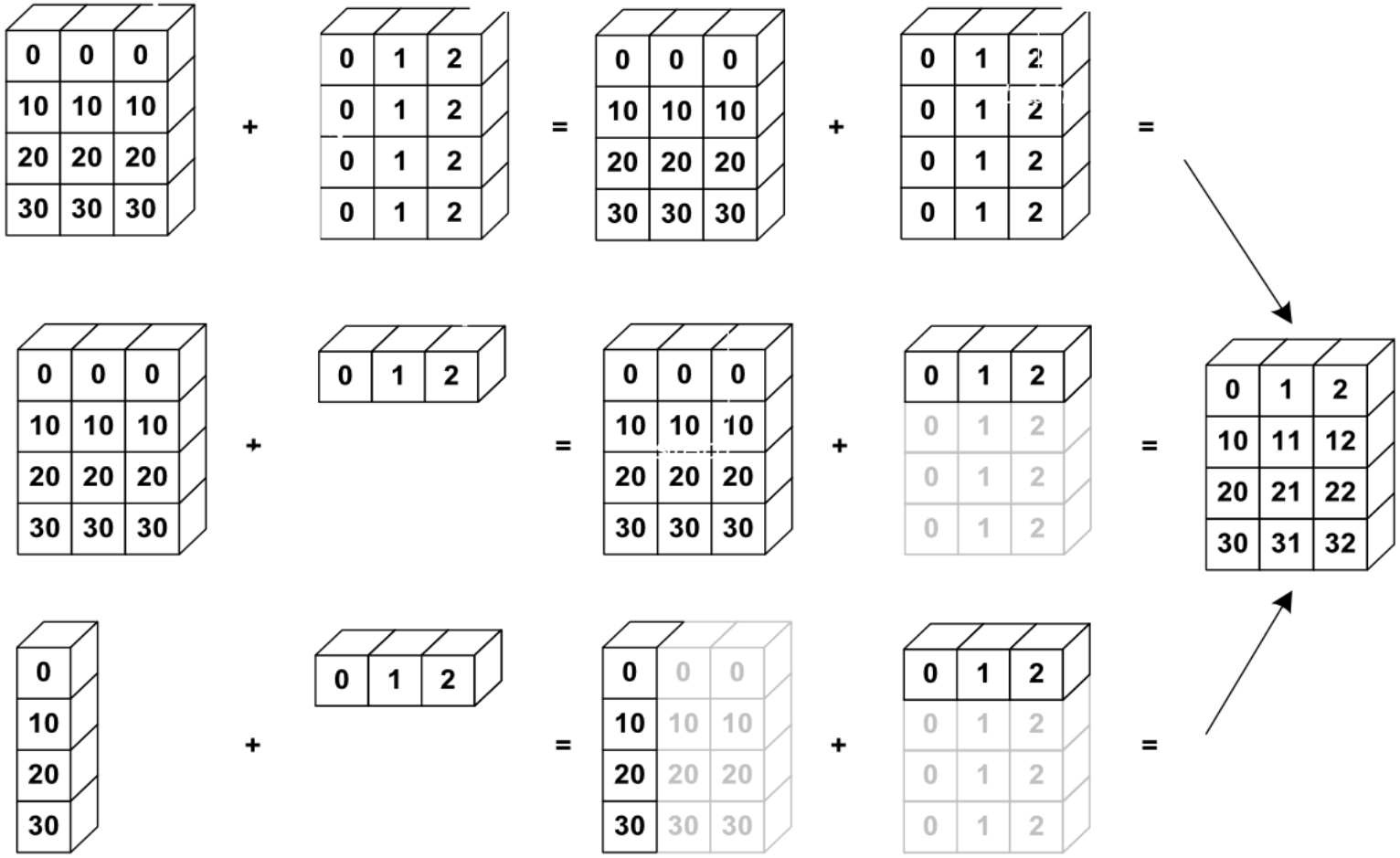
array_broadcasting .py

```
import numpy as np

# multiplication by a scalar
arr = np.random.random((4, 5))
result = arr * 5            # multiply each element by 5

# scales the first column by 0.
# scales the second column by 1.
# etc.
result = arr * np.arange(5)
```

Array Broadcasting



From Scipy Lecture Notes
(`http://www.scipy-lectures.org/index.html`)

Summary

- Know how to create arrays : array, arange, ones, zeros.
- Know the shape of the array with array.shape, then use slicing to obtain different views of the array: array[::2], etc. Adjust the shape of the array using reshape or flatten it with ravel.
- Obtain a subset of the elements of an array and/or modify their values with masks (boolean indexing)
- Know miscellaneous operations on arrays, such as finding the mean or max (array.**max**(), array.mean()).

  No need to retain everything, but have the reflex to search in the documentation

  (online docs, **help**(), lookfor()!!
- For advanced use:
  - Master the indexing with arrays of integers, as well as broadcasting.
  - Know more Numpy functions to handle various array operations.